Solution:

(i) A is translated to A' by 5→,2↑

(ii) B is translated to B' by 5←,3↑

(iii) C is translated to C' by 4←,5↓

DO YOU KNOW?

The sights and pageantry of marching band performance can add to the excitement of a sporting event. Band members dedicate a lot of time and energy to learning the music as well as the movements required for a performance.

The movements of each band member as they progress throughout the show are examples of *translations*.



4.2.2. Reflection

A ***reflection*** is a transformation that "flips" or "reflects" a figure about a line.

After a figure is reflected, it looks like a mirror image of itself. The line that a figure is flipped over is called ***a line of reflection.***

We can observe reflection in water, a mirror or in a glossy surface as shown in *Fig. 4.9*.

Reflection in Mirror



Reflection on glossy surface



Reflection in Water

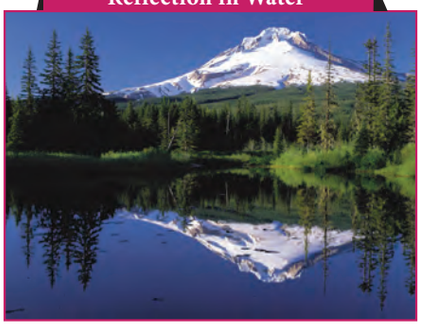


Fig. 4.9

Observe the following pictures.

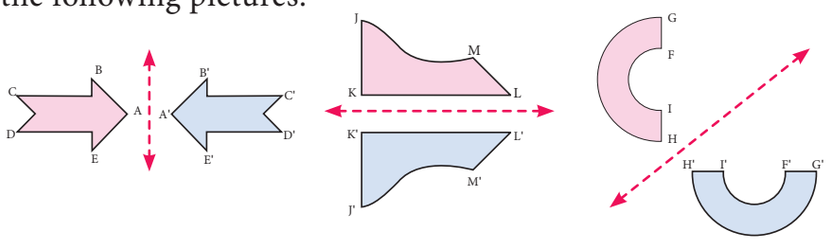


Fig. 4.10

In the above pictures (*Fig. 4.10*), the figures are reflected by a line. This line is called a line of reflection. Here the red line is the line of reflection.

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